Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Physicist (Ph.D. and M.Sc. in Physics) with ten years of research in computational astrophysics. Taught a graduate thermodynamics course and mentored three graduate researchers who co-authored papers with me. Teaching approach built on intuition, critical thinking, and collaborative problem-solving.

Teaching and Mentoring

Thermodynamics (graduate level) Jun 2018
Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Substitute Instructor

Zachary Davis, graduate student 2018 – 2022
Department of Physics and Astronomy, Purdue University Mentoring

Amanda O. Harrison, graduate student 2018 – 2020
Department of Physics and Astronomy, Purdue University Mentoring

Hao Zhang, graduate student 2018 – 2019
Department of Physics and Astronomy, Purdue University Mentoring

Education

Universitat de València, *Ph.D. in Physics, Excellent Cum Laude* Jul 2017

Universidad Michoacana de San Nicolás de Hidalgo, *M.Sc. in Physics* Sep 2011

Universidad Autónoma del Estado de México, *B.Sc. in Physics* Dec 2008

Experience

Independent Contractor, *Mercor* Jan 2026 – Present

- Evaluate physics reasoning in AI model outputs for technical accuracy, identifying failure modes and providing structured feedback.

Postdoctoral Research Fellow, *Purdue University* Oct 2018 – Nov 2020

- Obtained and led a NASA-funded grant studying radiation from active galaxies.
- Mentored three graduate researchers; two are now postdoctoral researchers.
- Published 3 peer-reviewed papers.

Postdoctoral Research Associate, *Rochester Institute of Technology* Feb 2021 – Apr 2022

- Led an NSF-sponsored project upgrading a high-performance simulation code for modeling black hole binaries.
- Collaborated on a multi-institutional study of binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow, *Universidad Michoacana de San Nicolás de Hidalgo* Jan – Sep 2018

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Outreach

Los más rápidos y los más furiosos (The Fastest and the Most Furious) Sep 2020
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco Online talk

Una simulación de la física y la astrofísica (A Simulation of Physics and Astrophysics) Aug 2020
Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco Online talk

Skills

Physics: Mechanics, electromagnetism, thermodynamics, relativity, statistical physics, astrophysics
Computing: Python, Fortran, C/C++, high-performance/parallel computing
Languages: English (fluent), Spanish (native), Nahuatl (basic)

Selected Publications (9 total, 3 first-author)

Full list available on request.